20. Path-integral thermodynamics with a transparent error budget

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

At small scales, particles cannot always be treated as tiny classical balls sitting at definite positions. Their quantum spreading affects thermal averages, especially for light particles. Path-integral methods replace part of this quantum calculation with a sampling problem involving an artificial ring of linked copies of a particle. This challenge checks that the sampling produces the right equilibrium quantities and that each source of error is reported separately.

Key terms

- **Thermodynamics** — The study of quantities such as energy and temperature describing equilibrium behavior.
- **Path integral** — A formulation of quantum mechanics involving a sum over possible paths; here it is used in imaginary time for equilibrium properties.
- **Imaginary time** — A mathematical reformulation of time used to connect quantum calculations with thermal statistics, not a physical clock reading.
- **Ring polymer and beads** — The artificial closed chain and its particle copies used to approximate the imaginary-time path integral.
- **PIMD** — Path-integral molecular dynamics, a method using auxiliary dynamics to sample the ring-polymer equilibrium distribution.
- **Estimator** — A formula calculated from sampled configurations to estimate a desired physical average.
- **Discretization bias** — Systematic error caused here by using a finite number of beads or a finite integration step.
- **Sampling uncertainty** — Uncertainty from using only a finite sample of configurations.
- **Born–Oppenheimer surface** — An approximate potential energy for nuclear motion obtained by treating electronic behavior separately.

Why this matters

Quantum thermal effects matter for understanding how light particles behave in molecules and materials. But an accurate-looking simulation can combine several hidden errors: too few beads, incomplete sampling, time-step effects and an approximate potential. A transparent error budget makes the calculation useful as a scientific measurement of the model rather than just a plausible plot.

What progress would mean

The initial result would reproduce known oscillator averages and then handle one controlled anharmonic case. It would validate equilibrium predictions; it would not automatically give exact real-time reaction rates or remove approximations in an electronic-structure model.

Core question

For a specified one-dimensional quantum Hamiltonian, can path-integral sampling reproduce equilibrium energy and position moments with separately assessed discretization and statistical errors?

Scope and assumptions

Start with H=p^2/(2m)+V(q), m>0, β>0, and a confining harmonic potential; extend to an explicitly defined anharmonic potential. Nuclei are distinguishable Boltzmann particles. Report only equilibrium observables initially.

An ab initio extension must specify the Born–Oppenheimer surface, electronic method, basis, and force errors. It is not exact quantum chemistry merely because no empirical force field is used. Real-time rates and exchange statistics require distinct methods.

Mathematical target

Set β_P=β/P and ω_P=P/($\beta\hbar$). Sample the coordinate density proportional to

$$\exp\left[-\beta_P \sum_{s=1}^{P} \{\tfrac{1}{2} m\omega_P^2 (q_s - q_{s+1})^2 + V(q_s)\}\right], \quad q_{P+1} = q_1.$$

Coordinate observables use P$^{-1}\Sigma$_s A(q_s), generally not A(q_centroid). For one dimension the primitive energy estimator is P/(2β)−mP/(2$\beta^2\hbar^2$)Σ_s(q_s−q_{s+1})^2+P$^{-1}\Sigma$_sV(q_s). Derive the chosen estimator from the partition function to fix all temperature factors.

Required outputs and proof obligations

Return algorithm, seeds, temperatures, bead counts, time-step settings, estimators, and an error ledger separating sampling, integration, finite-P, and potential/model errors. Use "statistical interval" for sampling uncertainty. Reserve "rigorous enclosure" for cases with proved sampling/discretization/tail bounds.

Validation and rejection controls

Recover E=($\hbar\omega$/2)coth($\beta\hbar\omega$/2) and $\langle$q$^2\rangle$=$\hbar$coth($\beta\hbar\omega$/2)/(2mω) for the oscillator over a declared β range. Compare multiple P and time steps, independent chains, and autocorrelation diagnostics. Validate anharmonic results against a separately converged or certified spectral method. A convergence fit alone cannot certify its asymptote.

Milestones and research extension

Baseline: oscillator thermodynamics within predeclared statistical and bias tolerances. Strong: one anharmonic error-controlled result. Research extension: a small ab initio system with a full error budget; rigorous certification requires more than SCF convergence and stable trajectories.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Ceriotti et al., Efficient stochastic thermostatting of path integral molecular dynamics — Normal-mode sampling and stochastic thermostat methods.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\beta_P = \beta/P, \quad \omega_P = P/(\beta\hbar), \quad q_{P+1} = q_1,$$

$$\pi_P(q) \propto \exp\left[-\beta_P \sum_{s=1}^{P} \left\{ \frac{m\omega_P^2}{2}(q_s - q_{s+1})^2 + V(q_s) \right\}\right],$$

$$E_{\text{prim}} = \frac{P}{2\beta} - \frac{mP}{2\beta^2\hbar^2} \sum_s (q_s - q_{s+1})^2 + \frac{1}{P}\sum_s V(q_s).$$

Coordinate observables use bead averages. Derive the estimator from the same partition-function convention. Separate sampling, integration, finite-P and model errors; convergence fits and empirical autocorrelation diagnostics do not by themselves give rigorous enclosures.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — inconsistent temperature/action (70–92).** The exact identity for $\exp(-\beta \mathrm{H})$ is confused with a Trotter approximation, and the action/ring-polymer formulas mix β and β/P. The revised density states one consistent convention.
2. **Blocking — estimators (110–120).** The original primitive energy omits the spring contribution, while nonlinear observables at the centroid do not equal bead-averaged observables. The oscillator provides a decisive regression benchmark.
3. **Major — proof claims (45, 56–59, 132–140).** P-extrapolation, energy conservation, and estimated autocorrelation time do not prove ergodicity or a rigorous total error bound. Statistical evidence is useful but must be labeled.
4. **Major — exactness and applications.** Finite-P PIMD with an approximate electronic surface has several approximations; equilibrium imaginary-time sampling does not directly certify real-time tunneling rates. The rewrite makes those limits part of the task definition.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a specified one-dimensional quantum Hamiltonian, can path-integral sampling reproduce equilibrium energy and position moments with separately assessed discretization and statistical errors?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: oscillator thermodynamics within predeclared statistical and bias tolerances. Strong: one anharmonic error-controlled result. Research extension: a small ab initio system with a full error budget; rigorous certification requires more than SCF convergence and stable trajectories.

Sources supporting the corrections

- Ceriotti et al., Efficient stochastic thermostatting of path integral molecular dynamics — Normal-mode sampling and stochastic thermostat methods.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.